Doxorubicin alters G-protein coupled receptor-mediated vasocontraction in rat coronary arteries

Caroline Lozahic, H Maddock, Mark Wheatley, Hardip Sandhu

- Centre for Health and Life Sciences
- Centre for Intelligent Healthcare

Research output: Contribution to journal › Article › peer-review

Abstract

Doxorubicin (Doxo)-associated cardio-and vasotoxicity has been recognised as a serious complication of cancer chemotherapy. The purpose of this novel paper was to determine the effect of Doxo on G-protein coupled receptor (GPCR)-mediated vasocontraction located on vascular smooth muscle cells. Rat left anterior descending artery segments were incubated for 24 h with 0.5 µM Doxo. The vasocontractile responses by activation of endothelin receptor type A (ETA) and type B (ETB), serotonin receptor 1B (5-HT1B) and thromboxane A2 prostanoid receptor (TP) were investigated by a sensitive

myography using specific agonists, while the specificity of the GPCR agonists was verified by applying selective antagonists (i.e. ETA and ETB agonist = 10-14-10-7.5 M endothelin-1 (ET-1); ETA antagonist = 10 µM BQ123; ETB agonists = 10-14-10-7.5 M sarafotoxin 6c (S6c) and ET-1; ETB antagonist = 0.1 µM BQ788; 5-HT1B agonist = 10-12-10-5.5 M 5-carboxamidotryptamine (5-CT); 5-HT1B antagonist = 1 µM GR55562; TP agonist = 10-12-10-6.5 M U46619; TP antagonist = 1 µM Seratrodast).
Our results show that 0.5 µM Doxo incubation of LAD segments leads to an increased VSMC vasocontraction through the ETB, 5-HT1B and TP GPCRs, with a 2.2-fold increase in ETB-mediated vasocontraction at 10-10.5 M S6c, a 2.0-fold increase in 5-HT1B-mediated vasocontraction at 10-5.5 M 5-CT, and a 1.3-fold increase in TP-mediated vasocontraction at 10-6.5 M U46619. Further studies unravelling the involvement of intracellular GPCR signalling pathways will broaden our understanding of the Doxo-induced vasotoxicity, and thus pave the way to mitigate the adverse effects by potential implementation of adjunct therapy options.

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Keywords

- G-protein coupled receptors
- Doxorubicin
- cancer therapy adverse effect
- vasotoxicity
- cardiotoxicity
- coronary microvascular function

UN SDGs

This output contributes to the following UN Sustainable Development Goals (SDGs)

- SDG 3 - Good Health and Well-being

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